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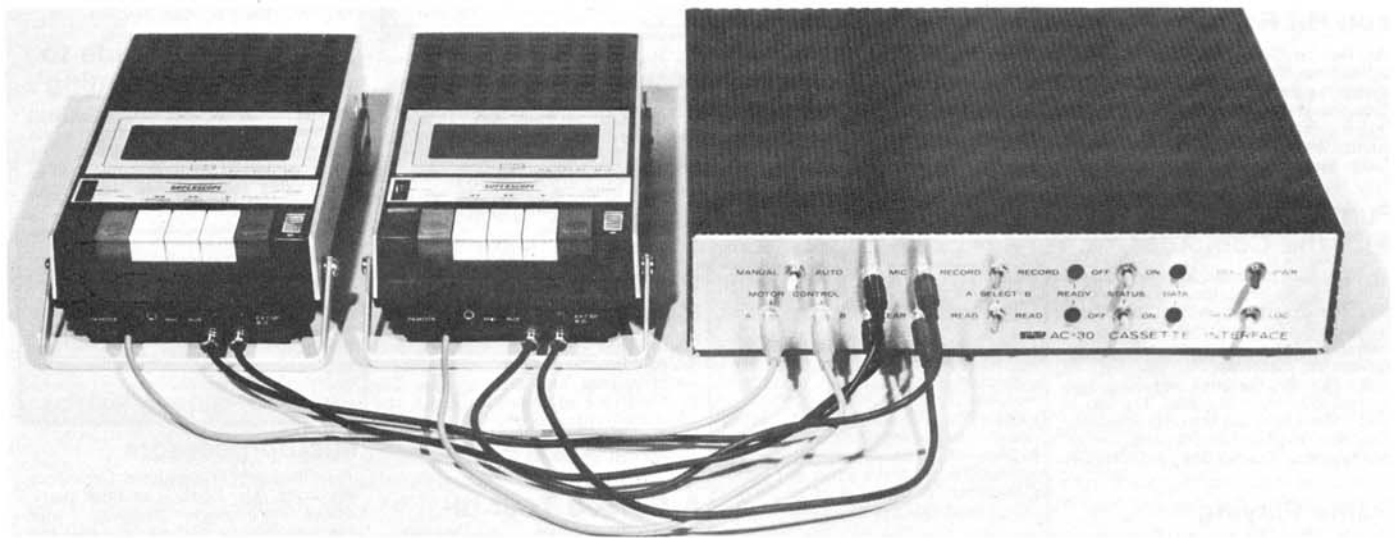


Photo 1: The AC-30 Cassette Interface Unit shown in its natural arrangement with a couple of inexpensive cassette recorders. The cables going from the AC-30 to the recorders can be fabricated at home from standard high fidelity cables cut in half, or complete cables can be purchased to match your recorder to phono style plugs. See your local Radio Shack or equivalent store for such items.

the AC-30

Gary Kay
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Cassette tape is one of the most flexible and least expensive means of mass data storage for computer systems. When compared to paper tape readers and punches, you'll find that although the paper tape readers can be made rather inexpensively, the punches cannot. Paper tape systems are typically slower and the punched tapes cannot of course be repunched and used over and over again, as you can with cassettes. Disk systems on the other hand offer significant advantages over cassettes but are still too expensive for many applications, and for most low cost personal computer users. Even those lucky enough to have a disk system still need a more universal medium for exchanging programs.

Although there are several commercial digital cassette tape decks on the market today, recording techniques vary, and they are of course much more expensive than the average audio cassette unit. As could be expected most hobbyist computer system

mass data storage designs have been based on the audio cassette recorder.

The use of inconsistent recording techniques among the various manufacturers makes it impossible, for example, to record a program, or data tape on a SWTPC 6800 computer system and play it back on a MITS 680 computer system. In order to coordinate manufacturer design efforts, and exploit the most effective recording technique, BYTE magazine sponsored a symposium in the fall of 1975 in Kansas City in an attempt to establish a recording standard for the storage of digital data on audio cassette recorders. The standard which was adopted has been tested and fully supported by Southwest Technical Products Corp. It appears to be the best compromise between economy and reliability. Although complete details are contained in the February and March 1976 issues of BYTE magazine, the recording philosophy is to record data serially using the standard UART format at

300 baud (30 characters per second). Marks or logic ones are represented by recording a 2400 Hz sine wave on the tape while spaces or logic zeros are represented by recording a 1200 Hz sine wave. With the proper circuitry this recorded data can then be read off the tape and converted into parallel data using a self clocking UART system which will tolerate audio recorder speed variations of approximately $\pm 30\%$. This figure is far better than that of most other modulation techniques and is a real advantage when you consider the degree of worst case speed variation between inexpensive audio recorders, in addition to which we have speed variations due to line voltage, battery voltage, wow and flutter, mechanism wear, etc. Thus evolved the "Kansas City" standard. It should be noted that the standard does not specify how the data is to be organized on the tape, so there can be, and probably will be some incompatibility among various manufacturer's software. This is a software problem which is fairly easy to resolve, given documentation of the data being sent.

Since the definition of the "Kansas City" standard, there have been several articles printed on circuits conforming to the standard, but there has yet to be a true audio cassette interface "system." When considering an audio cassette tape interface system, I think the potential user should ask the following questions:

1. Can the cassette interface be added to the computer system in such a way as to take full advantage of the computer system's existing resident tape load and dump routines?
2. Can the cassette unit be interfaced to the computer system without requiring the use of an additional interface port for the computer system?
3. Can the single cassette interface unit simultaneously or independently operate two audio cassette recorders (one reading while the other is re-

cording) and if so can the user simply switch select the function of each recorder instead of swapping a multitude of patch cords?

4. Will the cassette interface provide manual or computer control (switch selectable) over either cassette recorder's motor operation in both read and record modes?
5. Does the interface have status indicators to show read and record states as well as valid data flow?
6. Can the cassette interface unit simultaneously operate with a computer or a 300 baud terminal, switch selectable, allowing you to use your terminal in a stand alone mode to record or visually examine data on tapes before loading them into your computer?
7. Can the unit be tied to a 300 baud terminal like the TV typewriter II so as to respond to reader on, reader off, record on and record off control commands just like a teletypewriter with automatic reader/punch features?
8. Is the cassette interface unit complete with chassis, cover and 120/240 VAC, 50 to 60 Hz internal power supply?

Well, these are the questions I asked when I set about to design the SWTPC AC-30 audio cassette interface. This article contains my answers to these questions as built into the AC-30. Although it has been designed for and used extensively with the SWTPC 6800 computer system and CT-1024 (TV typewriter II) terminal system, the AC-30 has been designed to be as universal and flexible a system as possible. If your computer's control terminal is interfaced to the computer through a 300 baud, RS-232 compatible serial interfaces with accessible clocks at 16 times the data rate on both computer and terminal, the SWTPC AC-30 cassette interface unit is simply plugged between the computer and terminal inter-

faces. This is the ideal mode of operation since the cassette unit can take full advantage of computer resident tape load and dump routines and requires no additional interfaces. Switching the cassette unit to the local mode directly interconnects the terminal and cassette unit for terminal "only" cassette tape operation just like the local mode of operation on teletypewriters. While operating in the remote mode, the computer communicates with both the terminal and cassette unit, here again just like the remote mode of operation on teletypewriters. Those individuals using the SWTPC CT-1024 terminal system or any terminal system with accessible control character decoders may even pick reader on (control Q), reader off (control S), record on (control R), and record off (control T) control commands right off the control character decoder circuitry on their terminal system, giving the computer system program control over cassette recorder data flow and

even motor operation. Those not having access to decoded control commands may still have cassette control by driving the cassette interface with control lines from a separate parallel interface option located on the attached computer system.

Those users not operating their control terminal RS-232 serial at 300 baud or not having access to their terminal's UART clock (at 16 times the data rate) may still use the cassette interface, but must attach it to the computer system through a separate RS-232 serial 300 baud interface with accessible clocks located in the computer system. This however eliminates the ability to use the computer resident control terminal tape load and dump routines as well as the local/remote feature described previously.

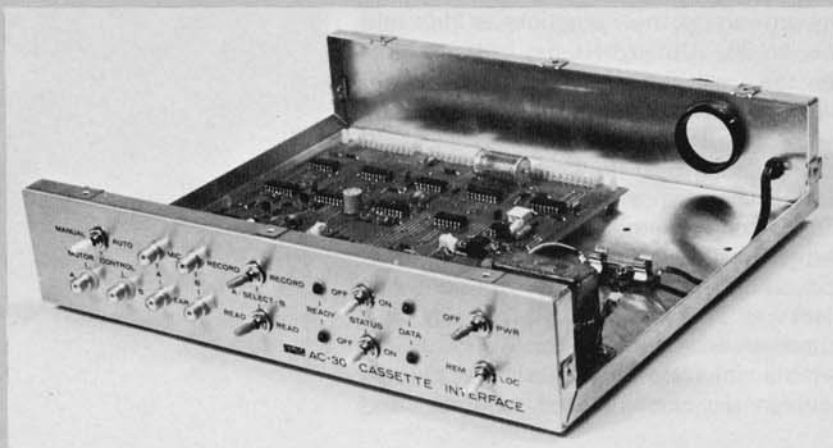
The cassette interface circuitry is constructed on a 7 3/4 by 7 1/2 inch (19.7 by 19.1 cm) double sided, plated through hole fiberglass circuit board with all electrical connections made to the board through one

Some Comments on the AC-30

by Carl Helmers, Editor

I purchased an AC-30 for my homebrew 6800 system when I stopped by at SWTPC for a visit following the ARRL national convention in Denver this past July. It took me two days (about 16 hours) on a weekend to put together this kit. My experiences confirmed Gary Liming's report in this issue. It was my first kit in recent memory; my last previous experience with kit assembly of electronics was a Heathkit "Benton Harbor Lunchbox" 2 meter amateur radio rig which I built with some help from experienced friends more than 10 years ago while in high school. The AC-30 went together at a "slow but steady" rate with the instructions serving as my guide.

Photo 2.



I had a big psychological problem (from bad experiences with some earlier CMOS circuits) convincing myself to actually solder a CMOS integrated circuit into place. The argument which Gary Kay, Joe Deres and Dan Meyer at SWTPC give for using direct soldering of components has two major points: first, their firm uses only new integrated circuits which have a demonstrated low probability of failure if inserted correctly; second, a hard soldered connection is much less likely to suffer a mechanical problem than a socket. A corollary of the socket problem is the temptation of novices to use "any old socket" available inexpensively, which results in connection problems which multiply the possible sources of error. I saw a couple of beauties waiting in the SWTPC repair queue: kits which had been returned because "they don't work" according to their owners, but which had sockets which looked like refugees from a salt water bath. (The repair technique? Throw out the board, and rebuild using the formerly socketed integrated circuits!) So, in assembling my unit, I crossed my fingers, grounded myself and the soldering iron tip with clip leads, and took Gary Kay's persistent advice to solder the circuits in place. The result is seen in photo 2. The only problem I had in checking out the unit was a mischievous solder bridge (due to my own carelessness) that didn't look like a bridge at

of the five edge connectors. The three connectors along the back edge of the circuit board are for connections to the computer, control decoder and terminal while the two along the front edge are for connections to the cassette interface's control panel. The printed circuit board in turn is mounted inside a 12 3/4 inch wide by 3 inch high by 11 inch deep (32 by 7.6 by 30 cm) aluminum chassis with a silver dress panel and black anodized perforated cover. The complement of front panel switches, indicators and jacks includes the following:

MIC, EAR and REMOTE jacks for recorder A: These jacks are connected through patch cords to the cassette recorder's respective jacks. It is often-times necessary to patch the MIC output of the cassette interface to the AUX input rather than the MIC input of the recorder. Some experimentation may be necessary here. Be sure the cassette recorder(s) you select have a

remote jack on them. This is necessary in order to have cassette recorder motor control, an essential part of convenient user software.

MIC, EAR and REMOTE jacks for recorder B: These jacks may be used for feeding a second cassette recorder, often required when using tape to tape software packages. Their functional description is identical to that provided for recorder A.

Record Select A or B: When this two position switch is in the A position, the cassette interface will output data to cassette recorder A. When in the B position it will output data to cassette recorder B.

Read Select A or B: When this two position switch is in the A position, the cassette interface will read data from cassette recorder A. When in the

all. Photo 3 shows the rear of the front panel in the unit I assembled.

Extending the AC-30's Usefulness

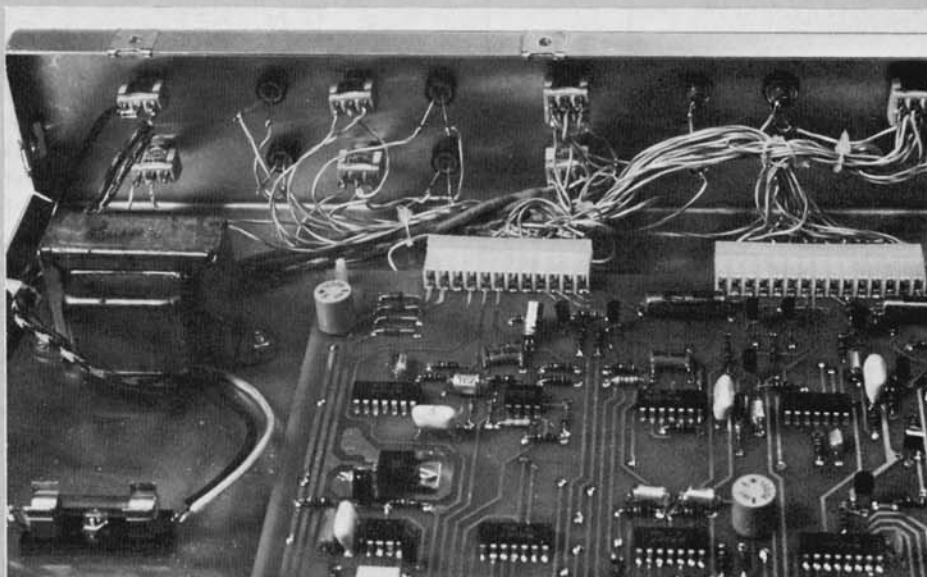
While I have not tried the following suggestion yet, in preparing this article on the AC-30 the following thought occurred: The circuit of figure 1 does not care if the UART is really running at 16 times the clock rate. By eliminating the redundancy of the "Kansas City" standard, and generating one's own signals with a programmed data formatting routine, it should be possible to use the AC-30 through a PIA port and RS-232 level shifters to run at 2400 baud. [Authors take note: Details of such a program to modify the AC-30's normal usage mode would make an excellent article.] If parallel IO and computation is required, multiprogramming using the processor's interrupt structure would be required, with the addition of an output clock generator to keep track of real time without programmed loops. (Input would drive an interrupt line directly from the clock extracted off the tape by the AC-30.)

Some Design Ruminations

In a preliminary calculation assuming for example my 6800 system with its current 833 kHz clock, the 4800 Hz data clock rate allows about 174 processor cycles for each clock period of the IO operation. Of these

cycles, at least 22 are required for the fixed overhead of a 6800 interrupt, whether or not the interrupt does anything (12 cycles to get to the interrupt handler, 10 to return). Assuming that this is the only active interrupt, the input or output data handler for each bit could conservatively be expected to take less than 100 additional cycles (20 to 30 instructions or so). Thus with an interrupt driven programmed interface, I might expect about 70% of real time to be occupied by IO during read or write operations as a reasonable worst case prediction of processor utilization. A similar calculation could be done for the 8080, TMS9900, or other microprocessors with which the AC-30 could be used in this mode of operation. ■

Photo 3.



The schematic diagram illustrates a carrier detector circuit, labeled as Figure 2. The circuit is divided into several functional blocks:

- Low Pass Active Filter:** Utilizes a 1/2 4558 Dual Op Amp (IC4A) to filter the input signal. It includes resistors R1 through R7 and capacitors C1 through C4.
- RC High Pass Filter:** Consists of a network of resistors (R8, R9, R10, R11) and capacitors (C5, C6) to filter out low-frequency components.
- Comparator:** Uses a 1/2 4558 Dual Op Amp (IC4B) to compare the filtered signal against a reference level. It includes a diode (D1) and a resistor (R12).
- Edge Detector:** Employs two 4070 inverters (IC3D, IC3C) to detect the edges of the signal. It includes a 4.7V Zener diode (D4) and a 100K resistor (R13).
- Missing Carrier Detector:** Uses a 4001 monostable multivibrator (IC2B) to detect the absence of a carrier signal. It includes a 2N5210 transistor (Q1), a 2N5210 transistor (Q2), and a 4070 inverter (IC3A).
- Squaring Toggle and Data Toggle:** Utilizes 4013 flip-flops (IC5B, IC5A) to square the signal and toggle the data output. They include various resistors (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19) and capacitors (C1, C2, C3, C4, C5, C6, C7, C8, C9).

The circuit is powered by a +5V supply and includes various resistors (R1 through R19), capacitors (C1 through C9), and transistors (Q1, Q2, Q3). The output of the circuit is labeled "CARRIER DETECT".

B position it will read data from cassette recorder B.

Record Status On or Off: This three position switch is normally left in the center position allowing computer program generated control commands to set the state of the record latch. Momentarily flipping the switch to the ON or OFF position will manually update the status of the record latch. Leaving the switch in either the on or off position will override computer program control entirely. An LED status indicator just to the left of this switch always shows the state of the record latch. The operation of the cassette interface as a function of the state of the record latch is dependent upon the setting of the motor control switch which is described in detail later.

Read Status On or Off: This three position switch is normally left in the center position allowing computer program generated control commands to set the state of the read latch. Momentarily flipping the switch to the on or off position will manually update the status of the read latch. Leaving the switch in either the on or off position will override computer program control entirely. An LED status indicator just to the left of the switch always shows the state of the read latch. The operation of the cassette interface as a function of the state of the read latch is dependent upon the setting of the motor control switch which is described in detail later.

Record Data Indicator: This LED indicator shows the transmission of valid data out of the cassette interface. It lights only when the record latch is on and logic zeros or spaces are being transmitted. This allows the operator to confirm that a tape dump is in progress when lit, since the null data marking output does not light the indicator.

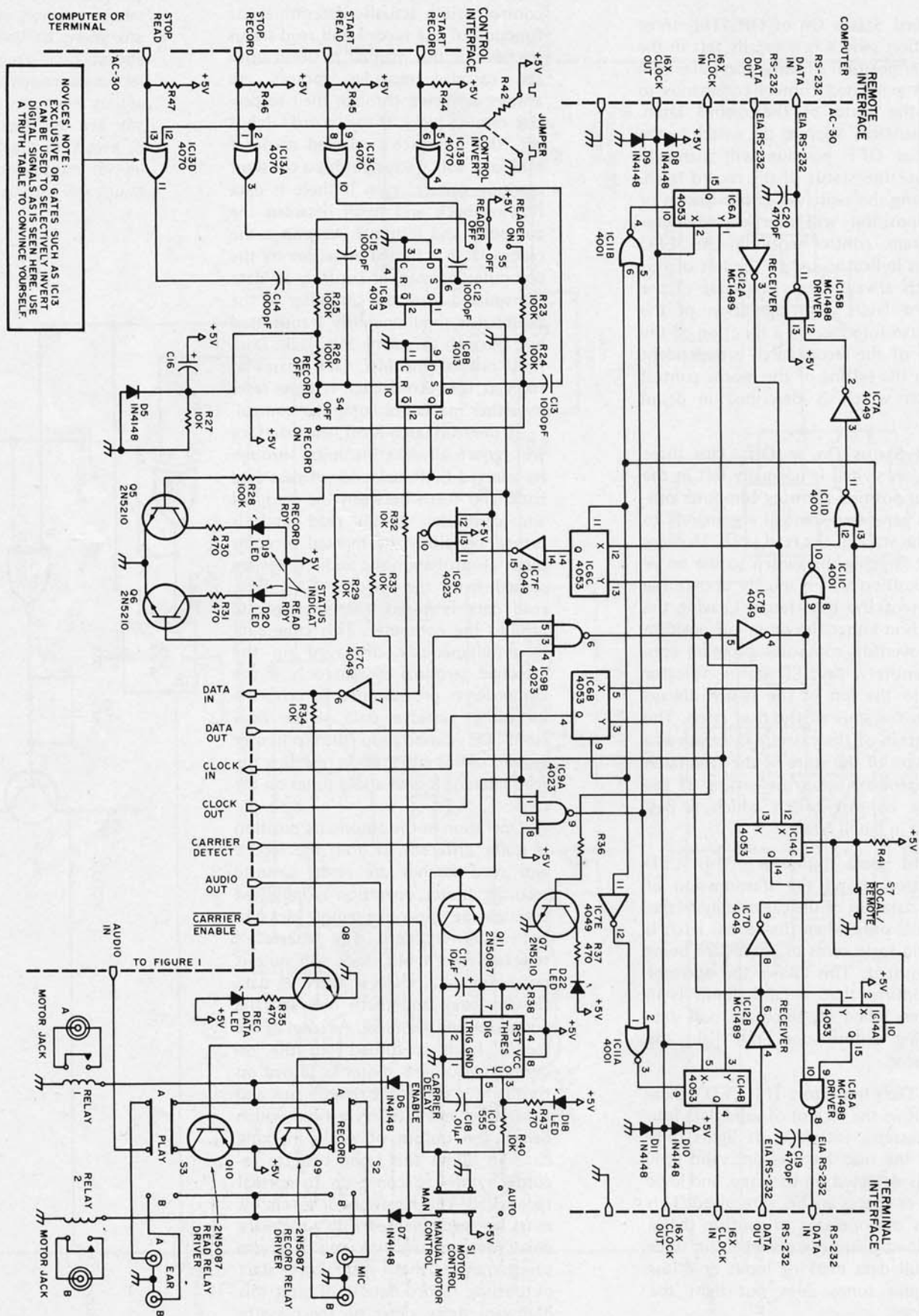
Read Data Indicator: This LED indicator shows the receipt of valid data into the cassette interface. It lights only when the read latch is on, valid FSK data is detected on the tape, and logic zeros or spaces are being received. This allows the operator to confirm that a tape load is in progress when lit, since the null data marking input or a loss of audio tones does not light the indicator.

Motor Control — Manual or Automatic: The position of the motor control switch actually determines the function of the record and read status latches. In the manual position both the cassette recorder motors are always activated through their respective remote jacks. If the record latch is off, the interface's selected recorder MIC jack will always output a constant marking carrier, even if there is data flowing back and forth between the computer and terminal. As soon as the record latch is turned on, either by the computer or manual control, all data transmitted from the computer to the terminal is simultaneously transmitted out through this same MIC jack. Data flow out of the MIC jack ceases as soon as the record latch is again reset by either manual or computer control.

If the read latch is off, the interface will ignore all data incoming through its selected EAR jack and yet pass data back and forth between the terminal and computer. If the read latch is turned on either by manual or computer control and valid audio tones are sensed from the selected EAR jack, read data is stored from the cassette unit to the computer. This same data is simultaneously displayed on the attached terminal system only if the computer is programmed to echo the incoming cassette data. Data flow from the cassette to the computer system ceases either upon resetting the read latch or loss of audio tones on the tape.

Operation in the automatic position is quite different. If both the record and read latches are reset, cassette recorder motor operation is inhibited through the respective remote jacks on both the recorders. The interface's selected record MIC jack will output no audio data, even if there is data flowing back and forth between the computer and terminal. As soon as the record latch is turned on, the recording recorder's motor is turned on through the respective remote jack and a variable delay timer is fired which delays the output of audio marking data to allow this same cassette recorder's tape to come up to normal tape speed. This hardware delay circuit must be supplemented with a software delay loop written into your programs to guarantee that you don't start outputting record data until after this hardware delay timer on the cassette

Figure 2: Control Logic of the AC-30. This diagram shows the computer and terminal interfaces of the AC-30.



interface has already timed out. When the record latch is again turned off, the interface will cease to output audio data and the selected recorder's motor is turned off. Here again it is wise to include a software delay loop in your programs to give the recorder time to come to a complete stop. This guarantees a sufficient gap between multiple recorded segments to allow one to do either incremental (start-stop) or continuous reads from the same tape.

When the read latch is turned on, the read recorder's motor is started. The interface inhibits all read recorder data until valid audio tones are detected, at which time all incoming cassette data is stored in the computer and simultaneously displayed on the terminal only if the computer's echo is enabled. Reads may be either continuous or incremental (start-stop). Since incremental tapes have blank gaps between recorded segments, the cassette interface's audio tone sensing circuitry has been designed to ignore all but the valid data segments stored on the tape.

Local/Remote switch: The local/remote switch on this cassette interface is analogous to that on standard teletypewriters. In the local mode there is a direct data link between the terminal and cassette recorder(s). The computer is electrically eliminated from the system. In the remote or normal mode of operation, the computer, terminal and cassette recorder(s) are all linked together.

Power On/Off: This switch controls AC power to the cassette interface unit. It must be powered up consistently with the interconnected computer and terminal systems even if cassette operation is not desired.

Attaching the Interface to the Recorders

If you will be using the interface just for loading and storing programs and data files to and from tape, you will probably need just one cassette recorder. If, however, you will be doing tape file editing or using assembler packages, you will probably have to use two cassette recorders. The same interface will handle both configurations. Electrical connections between the cassette interface and recorders are best made by cutting several 3 to 4 foot (90 to 122 cm) audio patch cords with the

molded RCA connectors on both ends, in half, and fitting the newly cut ends with the appropriate recorder jack mating connectors. Some cassette recorders such as the ones used with the prototype have both high level auxiliary and low level microphone inputs. In our case it was necessary to use the auxiliary input since the cassette recorder circuitry would not permit the reading of a tape with a plug simultaneously installed in the microphone jack. Be sure to use a cassette recorder featuring an AGC (automatic gain control) circuit. Most late model cassette recorders have this feature. Since cassette motor control is available, you will want a recorder with a remote jack that stops the recorder's motor. Here again recorder circuitry varies. Some units disable just the motor while others disable everything. It is best in this instance to just disable the motor. This will minimize the recorded transients between blocks of data when using the interface in the incremental (start-stop) mode of recording. The recorders used with the prototype were Superscope brand model C-101A [Superscope is a registered trademark of Superscope Inc.]. Although they were reliable, close examination of the quality of recorded data with an oscilloscope left a lot to be desired. The optimum volume control setting was around 7 on a scale of 0 to 10.

The quality of the cassette tape used with your recorder(s) will also affect the reliability of your system. Here the best rule of thumb is to assume you get what you pay for. The more expensive tapes will generally give better reliability. Remember too that you are recording audio frequency data, not saturated pulses. So don't use digital computer grade cassette tapes. Always erase previously recorded tapes with a bulk tape eraser such as a Radio Shack No. 44-210 (\$9.95) before rerecording them. This is especially important when using the system in the incremental (start-stop) mode of operation. Bulk tape erasers generally do a better job of erasing than a recorder's internal circuitry and guarantee that you don't pick up segments of previously recorded programs or data.

The interface's read circuitry must be calibrated for use in the read mode before using the interface. This is best done by first generating a calibration tape with continuous ASCII fives recorded on it and then reading back the tape to the terminal in the local mode while adjusting trimmer resistor R16 for a center setting between errored reads. ASCII fives have an alternating bit sequence ideal for calibration. It is a good idea to periodically recheck this setting using

your previously generated calibration tape.

Before using your cassette interface unit in the incremental (start-stop) mode of recording, it will be necessary to set the time delay on the carrier enable oneshot, trimmer resistor R39, so as to allow the recorder's motors to come up to speed before outputting a marking audio tone. This is best done by visually measuring your recorder's motor start time and then multiplying by two just for a safety factor. The interface's delay is then set by adjusting trimmer resistor R39 for this same time delay between the time the record "ready" indicator lights and the carrier enabled indicator (LED diode D18) comes on. The latter is mounted right on the interface's printed circuit board adjacent trimmer resistor R39.

Whenever you are writing to tape in the incremental mode you must provide a software delay loop in your program that is at least as long as the adjustable hardware timer delay plus 0.5 seconds. The additional half second is required to guarantee a carrier detect signal before data flow when the tape is being read. It is also a good idea to put a header character or characters ("synchronization" characters) at the beginning of each incremental record. Upon reading these records, your program should be written such that it ignores all data between these recorded segments until this header data is read. The cassette interface is more vulnerable to error reads between incrementally recorded data than at any other time. The end of each data block may easily be detected by recording some nondisplayed control character, or your software could include a block data count in its header. If you are using the cassette interface with the SWTPC 6800 computer system and CT-1024 (TV typewriter II) terminal system, the decoded record off ASCII control T command (hexadecimal 14) not only turns the recorder off, but is written at the end of the data record as well, for an end of record character.

Incrementally recorded records may be read either continuously or incrementally. In either case you must provide a software delay in your program at least one character time (33 ms) between the time the end of record character is read and the time data is output from the interface. The reason is that there is a slight delay between the loss of carrier on the tape and action of the carrier detect circuitry which in turn creates a gap in the 16 X UART clock. This gap causes no problems so long as you are not outputting data during the clock gap; thus the reason for the delay.

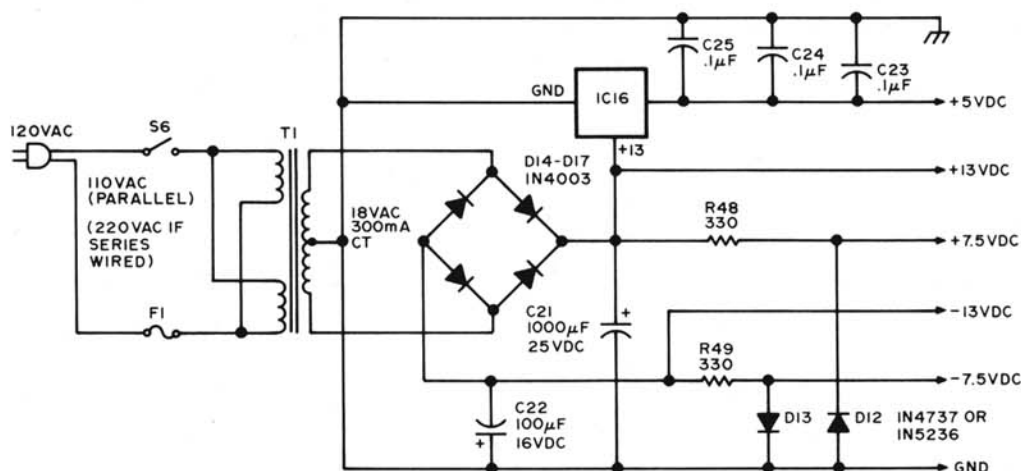
Fortunately consideration of tape gaps is

only necessary when using the interface in the incremental mode. Operating the system in the continuous mode isn't much different from paper tape operation. You will of course have to manually start and stop the recorders; but no special recording considerations are necessary, which usually allows use of unmodified computer resident tape load and dump routines for limited program storing and loading in a paper tape emulation mode.

Let's assume now we have a SWTPC 6800 computer and CT-1024 terminal system interfaced together with 300 baud RS-232 serial interfaces. As mentioned earlier for this configuration, the cassette interface can be connected in series between the two, taking full advantage of the computer's Motorola MIKBUG firmware tape dump and load routines. Let's also assume we have a program we wish to dump to tape which is already stored in the computer's memory with the computer's program counter, storage addresses (A048 and A049), set to the starting location of the program. Now use the memory change function to set the starting and ending addresses of the memory segment to be dumped in locations A002 through A005. The cassette interface's switches should be set for manual motor control with the record/read status switches in the center position and the local/remote switch in the remote position. Load a blank or previously erased tape into the selected cassette recorder and depress the rewind button to get to the beginning of the tape. Then depress the play button to advance several seconds beyond the leader on the tape. Stop the recorder and this time depress the cassette recorder record button. Then enter a P for punch on the terminal's keyboard. MIKBUG will interpret the P and then the record status light will come on, and the record data light will flicker as program data is stored to the cassette tape. When the dump is completed, both the status and data lights will go out. Do not stop the recorder; instead set memory locations A002 through A005 to dump the data stored in the program counter storage addresses as well. Since the record status light is not lit, none of this data passing back and forth between the terminal and computer is written to the still operating recorder. Again enter a P for punch. The status and data lights will again come on, but only for a fraction of a second since only two memory locations are written to the tape. Now while the cassette recorder is still recording, flip the local/remote switch to the local position, use the manual record status switch to set the record status latch and type in on the

The audio cassette interface described in this article is manufactured by Southwest Technical Products Corp, 219 W Rhapsody, San Antonio TX 78216. It is sold in kit form only, No. AC-30 for \$79.50, postpaid in the US. The kit includes circuit board, components, chassis, cover, power supply and assembly instructions but does not include the cassette recorders.

Figure 3: The power supply for the AC-30; see table 1 for wiring. Raw supplies of +13 V and -13 V are used by the EIA line drivers; zener regulated +7.5 and -7.5 are used by modem operational amplifiers; CMOS logic uses the +5 V supply.



terminal's keyboard S9 (the MIKBUG end of program code). Now you may stop the recorder. Don't forget to flip the local/remote switch back to the normal remote position.

Before trying to load this same program you might want to momentarily turn the computer off just to make sure the memory resident program is destroyed before loading. Reapplying power will force an auto-reset and start up the resident Motorola MIKBUG firmware on the SWTPC 6800. Rewind the tape to the beginning, making sure the cassette recorder's volume is set to a reliable setting, and depress the play button. Then type in a L for load on the terminal's keyboard. The read status light will immediately come on since the firmware outputs a read on control command. The data light however will remain off until program data is detected on the tape at which time it will begin to flicker. When it stops flickering it means that the program data has been loaded to the computer's memory but don't forget that the program counter data was also written to the tape, so the next flicker is that of the program counter being loaded. An S9 was also written to the tape; it is an end of tape marker. Upon reading this, the computer will output a read off control command forcing the read status light to go out, hence ignoring all subsequent data stored on the tape should the recorder be left to run. The program as well as the program counter have been loaded. Simply typing in the character G for go should initiate the program.

How It Works

For simplicity the cassette interface's circuitry has been broken up into three

separate systems: The modulator demodulator circuitry, the switching circuitry and the power supply. These are shown in figures 1, 2 and 3, respectively.

The modulator works by feeding a 4800 Hz (16 times 300 Hz) clock into the toggle provided by IC5b. The division by 2 in this flip flop insures a 50% duty cycle required by the modulator. The carrier enable input provides a means of suppressing audio output from the modulator. IC5a divides the frequency by two once more if the data in line is high and simply follows the clock frequency if the data in line is low. This gives a 1200 Hz tone for a low state and a 2400 Hz tone for a high state. The resulting output is then fed into two pole active filters provided by the 4558 operational amplifier section IC4a, where it is converted to a closer approximation of a sinusoidal audio waveform which is more easily handled by audio recorders.

Incoming audio data is first fed into a high pass filter consisting of R5 and C5 and then onto comparator IC4b. The compara-

Table 1: Integrated Circuit Power List. The supplies are taken from the circuitry in figure 3, and wired to the appropriate pins of these circuits.

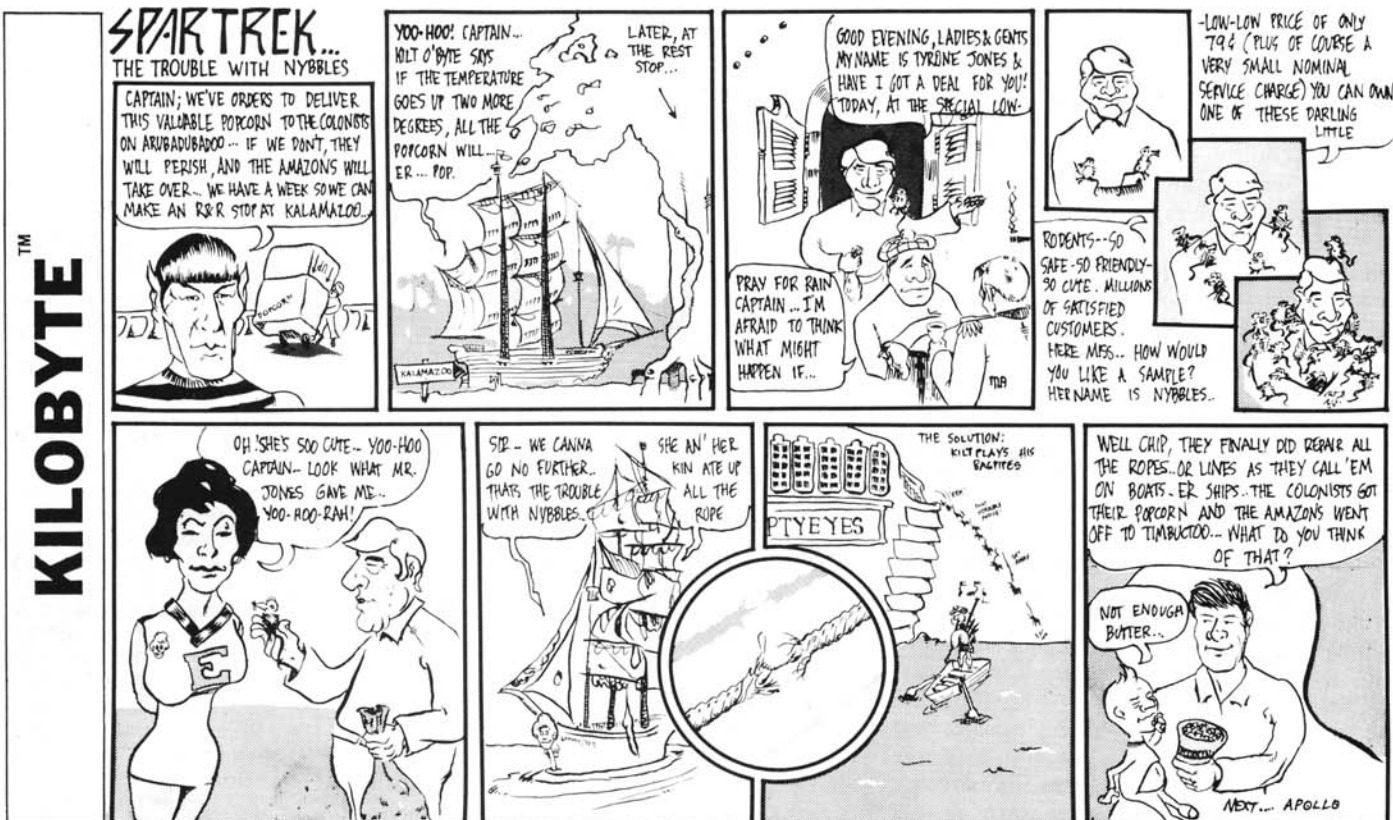
Number	Type	+5 V	GND	+7.5 V	-7.5 V	+13 V	-13 V
IC1	4013	14	7	—	—	—	—
IC2	4001	14	7	—	—	—	—
IC3	4070	14	7	—	—	—	—
IC4	4558	—	—	8	4	—	—
IC5	4013	14	7	—	—	—	—
IC6	4053	16	8	—	—	—	—
IC7	4049	16	8	—	—	—	—
IC8	4013	14	7	—	—	—	—
IC9	4023	14	1	—	—	—	—
IC10	555	8,4	1	—	—	—	—
IC11	4001	14	7	—	—	—	—
IC12	1489	14	7	—	—	—	—
IC13	4070	14	7	—	—	—	—
IC14	4053	16	8	—	—	—	—
IC15	1488	—	7	—	—	14	1

tor's 0.5 V hysteresis reduces the possibility of false triggering. The ± 7.5 V zener regulated power buses feeding IC4 are necessary to stabilize this hysteresis value. Zener diode clamp D3 limits the comparator output to CMOS compatible voltage levels. Each time the comparator changes states the exclusive OR functions of IC3c and IC3d generate a 5 μ s active low pulse at IC3 pin 11. This pulse is repetitive when data is being received and pulls capacitor C7 down to ground through diode D4. The level on capacitor C7 is in turn inverted and buffered through IC3a to form the carrier detect signal. This same active low pulse is inverted by IC2a where it feeds four separate circuits. The first is a missing pulse detector composed of transistor Q2 and inverter IC2b. This forces the buffered carrier detect signal low whenever several cycles of audio carrier are missed. The second circuit is also an adjustable missing pulse detector but this one times out whenever 1200 Hz data is being fed into the demodulator. The 20 k ohm trimmer resistor R16 sets the period for this timer. The third circuit driven by this pulse is the clock input to flip flop 4013 IC1a which outputs the demodulated data. This data out line is high when 1200 Hz audio is being demodulated and low when 2400 Hz audio is being demodu-

lated. The fourth circuit fed by the pulse is IC2d which with the addition of the output of IC3b synthesizes the 16 X clock out data. Take note that although on the average this clock is accurate, it jitters by design.

Within the control portion of the circuit, figure 2, integrated circuit halves IC8a and IC8b form status latches for the record and read circuitry. Either latch may be independently set or reset through manual toggle switches on the interface's front panel or through control pulses buffered and selectively inverted by IC13. Front panel LED indicators always show the current status of these latches.

The multitude of data and clock switching is performed by six data selectors within IC6 and IC14. NAND gates IC9a and IC9b inhibit data and clock flow out to the cassette recorders unless the read latch is set and valid recorder carrier data is detected. Timer IC10 provides the adjustable CARRIER ENABLE delay needed when using the system for recording in the incremental mode. All 300 baud data communication with the cassette interface is done through integrated EIA RS-232 receiver and transmitter integrated circuits IC12 and IC15 respectively. Motor control for each recorder is provided by reed relays RLY1 and RLY2. ■



Building the AC-30

Cassette Interface

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Any hobbyist who has ever loaded a sizeable program on a bit by bit or byte by byte basis, only to have it wiped out when the power is removed, really understands the significance of mass storage. Cassette recorders are attractive mass storage devices to a hobbyist because they are inexpensive and can be interfaced easily.

Southwest Technical Products Corp (219 W Rhapsody, San Antonio TX 78216) introduced their answer to mass storage with the AC-30 cassette interface in May 1976. *[See Gary Kay's article in this issue.]* At that time I had a 6800 processor using Motorola's MIKBUG as a monitor ROM and a CT-1024 with keyboard. While reading the advertisement brochure I was struck with fact that the interface made use of an already existing standard, the RS-232 interface. If I decided to try a different system later on, I would be able to keep the interface and tapes and use an easily acquired RS-232 connection on the new system. It also provided for two cassette drives for future system expansion. Seeing no problems connecting the AC-30 to my system, I ordered the AC-30 by phone for \$79.50 postpaid, and received it by UPS two weeks and two days later.

When the kit arrived, I immediately started studying the documentation. As I was reading, the directions mentioned a calibration program that should have been

supplied but was not included. I finished reading the instructions, made sure the parts were all there, and called SWTPC.

I spoke with an engineer who confirmed the missing page, and took my name and address. Only two days later a complete set of documentation appeared in my mailbox. The two days were not wasted, however, because it took me nearly that long to clear off a space on my workbench to build it!

The documentation package consisted of 22 pages of assembly instructions, complete schematics, a parts list, and a two color printed circuit board component layout. Included are two 17 3/4 inch by 22 1/2 inch (45 cm by 57 cm) charts showing wiring connections inside the AC-30 and between the terminal and computer system. Also, 11 pages of diagnostic and calibration software written for a 6800 system are included.

Assembly began with the 7 3/4 inch by 7 1/2 inch (19.7 cm by 19.1 cm) printed circuit board that I inspected, found no faults with, and which generally looked like a high quality board. Instructions for the kit are at a level of "Place all of the resistors on the board using the parts list and component layout drawing...", so you must at least know resistor color coding and be able to identify the parts. Using reference charts from electronics "how to do it" books will help to prevent mistakes for the novice.

Placing and soldering the 39 resistors, 25 capacitors, 16 integrated circuits and 11 transistors was time consuming, but not difficult. The soldering pads are placed well apart from each other, making soldering

easier than on some memory or processor boards. Solder is included.

Although there is a controversy among manufacturers, hobbyists, and engineers over the use of sockets, I decided to use them. Kit manufacturers don't advise sockets because they make a returned board very difficult to repair if the fault is due to a socket. A bad solder flow under the socket is hard to detect without tearing out the socket. Hobbyists sometimes get low quality sockets which can cause intermittent problems that are nightmares to debug.

I used quality sockets and verification soldering techniques because they help reduce integrated circuit handling (almost all of the integrated circuits in the kit are CMOS) and mainly because I would be doing my own repair work in the future. However, if you do not intend to repair your own equipment, there is little benefit in using sockets in a kit.

After completion of the printed circuit board, chassis assembly began. The 12 3/4 by 11 by 3 inch (32 cm by 30 cm by 7.6 cm) aluminum chassis holds the printed circuit board, transformer, fuse, and a dress front panel. A perforated black finish aluminum cover and rubber feet are included. All wiring to the printed circuit board is made through connectors which make removal of the board simple. The large connection charts are very helpful, and wire and wire ties are provided. Before starting chassis assembly, a little forethought about the routing of wire between the switches, phono jacks, LEDs, and printed circuit board connectors will help to save mistakes.

Front panel assembly went smoothly except for snapping the LED retainer clips into place. I found that using two ballpoint pen halves to push the clips together saved wear and tear on my fingernails.

Interconnecting cables between the terminal and computer are routed through a grommet directly to the printed circuit board connectors. In addition to the front panel switches, control inputs are provided for remote or computer control of the reader on, reader off, record on, record off, and Local/Remote modes. If you are using the CT-1024, these inputs can be controlled by using the CT-CA (computer controlled cursor board) which decodes various ASCII control characters that appear on the RS-232 interface. Otherwise, the controls can be operated via computer commands by building a decoder for the control characters or by dedicating a separate output port or address space for these functions. The interface requires both the terminal and computer to provide a X16 clock.

When assembly was completed, I bought cassette connecting cables (not supplied) and, like most recorders, mine required two miniature to phono and one subminiature to phono cables. I am using a Sankyo ST-50 recorder, picked out solely because it was the most inexpensive (\$44.95) recorder I could find that had automatic level control, auxiliary and remote inputs, and a tape counter. The tape counter is not necessary, but does allow you to easily catalog a number of programs on a single tape.

After connecting the recorder, two calibration steps are required. One is used to set the time between when the record on signal is received and when the record carrier comes up. This is necessary to allow the cassette motor to come up to speed before the recording begins, and is noncritical. The easiest way to adjust the delay is to visually inspect the start up time of the cassette motor, and set the delay greater than that. An on board LED is provided to indicate the carrier enable signal, and the adjustment is made with an on board pot and 555 timer.

The other adjustment is made to set the character frequency of the reader circuitry, and is done by loading the calibration program. This program outputs a control character to turn the recorder on, an ASCII five (chosen because of its alternating bit pattern), and a recorder off character, all in a continuous loop. After recording a tape of fives, the tape is then read back in local mode to the terminal. The calibration pot is then adjusted to center between the points at which errors occur. I found this to be a very wide range in which the reader would operate with no errors. This program is also useful in verifying new tapes for data use. A little experimentation showed that the best volume setting was about 7 on a scale of 0-10. The tone control seemed to make little difference except at extreme settings.

Now that calibration was completed, I loaded a program byte by byte for the last time, and dumped it out to my first mass storage peripheral! The program was approximately 2 K bytes and took 3 1/2 minutes to load back into the system. This is longer than 30 cps would account for, but is due to the Motorola MIKBUG dump and load routines that output two characters per byte (hexadecimal) and includes some error checking and addressing overhead.

The recording method is compatible with the "Kansas City" standard, and I have successfully read tapes generated by the "Bit Boffer" interface as described in the March 1976 issue of BYTE [page 30]. If there are any other 6800s with MIKBUG listening out there, drop me a card and let's swap!■